Fiscal Year:	FY 2009	**Task Last Updated:**	FY 12/31/2009
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNC06AA32A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan (University of Hawaii at Manoa) Sunderland, Peter (University of Maryland) Urban, David (NASA Glenn Research Center)		
Grant/Contract No.:	NNC06AA32A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This effort is a collaborative effort between the PI, the three Co-I's, and staff at NASA Glenn Research Center. The division of labor is roughly broken up with DLU and PBS performing microgravity experiments at the NASA Glenn Research Center, PBS directing numerical simulations at University of Maryland, BHC performing analytical work at University of Hawaii, and RLA performing normal gravity experiments at Washington University. The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The		

	proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed "Flame Design" as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	The following is a summary of the research activities during the period Oct. 2006 - Sept. 2009. 1) Numerical and Experimental Observations of Spherical Diffusion Flames. Spherical diffusion flames supported on a porous sphere were studied numerically and experimentally. Experiments were performed in 2.2 s and 5.2 s microgravity facilities. Numerical results were obtained from a Chemkin-based program. The program simulates flow from a porous sphere into a quiescent environment, yields both steady state and transient results and accounts for optically thick gas-phase radiation. The low flow velocities and long residence times in these diffusion flames lead to enhanced radiative and diffusive effects. Despite similar adiabatic flame temperatures, the measured and predicted temperatures varied by as much as 700 K. The temperature reduction correlates with flame size but characteristic flow times and Lewis number also influence temperature. The numerical results show that the ambient gas Lewis number would have a strong effect on flame temperature if the flames were steady and nonradiating. For example, a 10% decrease in Lewis number would increase the steady state flame temperature by 200 K. However, for these transient, radiating flames the effect of Lewis number is small. It was also observed that when hydrocarbon fuel is supplied from the ambient the large diffusion distances associated with these flames can lead to unusual steady state compositions near the outer boundary because decomposition products can diffuse to the outer boundary. This results in a loss of chemical enthalpy from the system but the effect on flame temperature is small. Transient predictions of flame sizes are larger than those observed in microgravity experiments. Close agreement could not be obtained without either increasing the model's thermal and mass diffusion properties by 30% or reducing mass flowrate by 25%. 2) Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity. Radiative extinction of spherical diffusion flames was investigated experimentally and numerically. The experiments involved microgravity spherical diffusion flames burning ethylene and propane at 0.98 bar. Both normal (fuel flowing into oxidizer) and inverse (oxidizer flowing into fuel) flames were studied, with nitrogen supplied to either the fuel or the oxygen. Flame conditions were chosen to ensure that the flames extinguished within the 2.2 s of available test time; thus extinction occurred during unsteady flame conditions. Diagnostics included color video and thin-filament pyrometry. The computations, which simulated flow from a porous sphere into a quiescent environment, included detailed chemistry, transport, and radiation and yielded transient results. Radiative extinction was observed experimentally and simulated numerically. Extinction time, peak temperature, and radiative loss fraction were found to be independent of flow rate except at very low flow rates. Radiative heat loss was dominated by the combustion products downstream of the flame and was found to scale with flame surface area, not volume. For large transient flames the heat release rate also scaled with surface area and thus the radiative loss fraction was largely independent of flow rate. Peak temperatures at extinction onset were about 1100 K, which is significantly lower than for kinetic extinction. An important observation of this work is that while radiative heat losses can drive transient extinction, this is not only because radiative losses are increasing with time but also because the heat release rate is falling off as the flame expands away from the burner and the reactant supply to the flame decreases. 3) Soot Inception Limits in Laminar Diffusion Flames with Application to Oxy–Fuel Combustion. For diffusion flames, the combination of oxygen enrichment and fuel dilution results in an increase in the stoichiometric mixture fraction, Zst, and alters the flame structure, i.e., the relationship between the local temperature and the local gas composition. Increasing Zst has been shown to result in the reduction or even elimination of soot. In the present work, the effects of variable Zst on soot inception are investigated in normal and inverse coflow flames, using ethylene as the fuel. Use of the inverse coflow flame underscores the validity of these concepts, since the convective field in the inverse flame directs particles into the fuel-rich region. Sooting limits based on particle luminosity are measured as a function of Zst. The sooting limit is obtained by varying the amount of inert gas until soot appears above a predefined height. For each limit flame, the adiabatic flame temperature is calculated and the flame temperature at the half-height is measured. The flame temperature at the sooting limit is found to increase with Zst for both normal and inverse flames. The effects of residence time are also investigated, and the sooting limit inception temperature is found to be dependent on fuel stream velocity for both the normal and inverse configurations. A simple model applicable to oxy–fuel combustion is presented which describes how increasing Zst results in the reduction and ultimately elimination of soot. This model assumes that soot inception can only occur in a region where critical values for species, temperature, and residence time are met. The soot inception region is shown to be bounded by two isotherms: a low-temperature boundary that is a function of residence time, and a high-temperature boundary that corresponds to the location of a critical local carbon-to-oxygen ratio. The effect of increasing Zst is to move the boundaries of the soot inception zone towards each other, until the zone is infinitely thin and thus the sooting limit is reached. By comparing the model to experimental data, a critical local C/O ratio of 0.53 and a sooting limit inception temperature of 1640 K (for a characteristic residence time of 22 ms) were determined for ethylene. 4) Characteristics of Non-Premixed Oxygen-Enhanced Combustion: I. The Presence of Appreciable Oxygen at the Location of Maximum Temperature. The presence of appreciable molecular oxygen at the location of maximum temperature has been observed in non-premixed oxygen-enhanced combustion (OEC) processes, specifically in flames having a high stoichiometric mixture fraction (Zst) produced with diluted fuel and oxygen-enrichment. For conventional fuel–air flames, key features of the flame are consistent with the flame sheet approximation (FSA). In particular, the depletion of O2 at the location of maximum temperature predicted by the FSA correlates well with the near-zero O2 concentration measured at this location for conventional fuel–air flames. In contradistinction, computational analysis with detailed kinetics demonstrates that for OEC flames at high Zst: (1) there is an appreciable concentration of O2 at the location of maximum temperature and (2) the maximum temperature is not coincident with the location of global stoichiometry, O2 depletion, or maximum heat release. We investigate these phenomena computationally in three non-premixed ethylene flames at low, moderate, and high Zst, but with equivalent adiabatic flame temperatures. Results demonstrate that the location of O2 depletion occurs in the vicinity of global stoichiometry for flames of any Zst and that the presence of appreciable O2 at the location of maximum temperature for high Zst flames is caused by a shift in the location of maximum temperature relative to the location of O2 depletion. This shifting is attributed to: (1) finite-rate

	multi-step chemistry resulting in exothermic heat release that is displaced from the location of O2 depletion and (2) the relative location of the heat release region with respect to the fuel and oxidizer boundaries in mixture fraction space. A method of superposition involving a variation of the flame sheet approximation with two heat sources is shown to be sufficient in explaining this phenomenon.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
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Articles in Peer-reviewed Journals	Skeen SA, Yablonsky G, Axelbaum RL. "Characteristics of non-premixed oxygen-enhanced combustion: I. The presence of appreciable oxygen at the location of maximum temperature." Combustion and Flame 2009 Nov;15(11);2145–52. http://dx.doi.org/10.1016/j.combustflame.2009.07.009 , Nov-2009
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Dissertations and Theses	Skeen SA. "Ph.D. Dissertation." Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University, 2009. , May-2009
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